

Electronic conductivity of solid electrolytes causes physical self-discharge in all-solid-state batteries

Received: 6 February 2023

Accepted: 20 May 2026

Published online: 17 August 2026

 Check for updates

Changhong Wang^{1,2,7} , Rong Xu^{3,7} , Yu Zhong^{4,7}, Han Su² , Suzhe Liang¹, Haoqi Ren², Jung Tae Kim² , Huan Huang⁵, Bing Xiao⁶ , Longan Jiao⁶, Jiangping Tu⁴  & Xueliang Sun^{1,2} 

All-solid-state batteries (ASSBs) with inorganic solid-state electrolytes (SSEs) hold vast potential for next-generation electric vehicles (EVs) due to their high energy density and enhanced safety. However, their self-discharge behaviour, a critical factor for EVs, has not been adequately investigated so far. Here we reveal that the electronic conductivity of SSEs, typically in the range of 10^{-8} – 10^{-9} S cm⁻¹, contributes to considerable physical self-discharge in ASSBs, particularly when the SSE thickness is on the order of tens of micrometres. To mitigate this physical self-discharge, the electronic conductivity of SSEs needs to be reduced to approximately 10^{-12} S cm⁻¹. However, none of the prevalent SSEs meets this threshold. Therefore, reducing physical self-discharge in ASSBs will require the development of SSEs with lower electronic conductivity, alongside rational interface and full-cell designs in future research. This study provides critical insights into the self-discharge phenomenon of ASSBs, which may reshape their future design and development.

All-solid-state batteries (ASSBs) have emerged as promising next-generation energy storage technology because of their enhanced safety, high theoretical energy density and cost competitiveness^{1–5}. As a critical material of ASSBs, solid-state electrolytes (SSEs) have been intensively developed, including sulfides⁶, oxides^{7,8}, halides^{9–13}, anti-perovskites¹⁴ and borohydrides^{15,16}, some of which exhibit ionic conductivities even higher than those of commercial liquid electrolytes¹⁷. With the advances in SSEs, ASSBs based on SSEs have demonstrated satisfactory electrochemical performance thus far, such as high-power ASSBs using superionic sulfide conductors¹, high-energy long-cycling all-solid-state lithium metal batteries with silver–carbon composite anodes², carbon-free high-loading silicon anodes using sulfide SSEs¹⁸,

and high-areal-capacity, long-cycle-life 4 V ceramic ASSBs enabled by halide SSEs¹⁹. These advancements strongly suggest ASSBs are the most promising technology for large-scale energy storage systems, such as electric vehicles (EVs) and grid-scale energy storage facilities. In these applications, ASSBs will be held at a high charge state for extended periods, such as when EVs are stored in a warehouse before delivery and parked in a garage or when grid-scale energy storage facilities hold electricity until demand. These scenarios set a stringent requirement for the self-discharge rate of ASSBs. However, the self-discharge behaviour in ASSBs has been neglected and unexamined thus far.

Self-discharge refers to the spontaneous consumption of stored energy in a battery under open-circuit conditions, in the absence of an

¹Eastern Institute for Advanced Study, Zhejiang Key Laboratory of All-Solid-State Battery, Ningbo Key Laboratory of All-Solid-State Battery, Ningbo Institute of Digital Twin, Eastern Institute of Technology, Ningbo, China. ²Department of Mechanical and Materials Engineering, University of Western Ontario, London, Ontario, Canada. ³State Key Laboratory for Strength and Vibration of Mechanical Structures, Department of Engineering Mechanics, Xi'an Jiaotong University, Xi'an, China. ⁴State Key Laboratory of Silicon Materials, Key Laboratory of Advanced Materials and Applications for Batteries of Zhejiang Province, School of Materials Science and Engineering, Zhejiang University, Hangzhou, China. ⁵GLABAT Solid-State Battery Inc., London, Ontario, Canada. ⁶Carl Zeiss (Shanghai) Co., Ltd., Shanghai, China. ⁷These authors contributed equally: Changhong Wang, Rong Xu, Yu Zhong.

 e-mail: cwang@eitech.edu.cn; xsun@eitech.edu.cn

external electrical load. Depending on the cause, self-discharge can be classified into physical self-discharge and chemical self-discharge. Physical self-discharge is influenced by the electronic conductivity of separators (in the case of ASSBs, this is the SSE) and the battery's state of charge. Chemical self-discharge, meanwhile, is mainly caused by spontaneous chemical reactions between the electrodes and electrolytes, as well as ion redistribution within the electrodes, leading to non-faradaic charge consumption and a decrease in the open-circuit voltage (OCV). Chemical self-discharge is typically irreversible, self-limited and strongly influenced by storage temperature and humidity²⁰.

In this study, we first perform a comparative analysis of the electronic conductivity of inorganic SSEs commonly used in ASSBs against commercialized polymer separators and lithium phosphorus oxynitride (LiPON), which are used in commercial lithium-ion batteries (LIBs) and solid-state thin-film batteries, respectively. This comparative analysis indicates that current inorganic SSEs have non-negligible electronic conductivity, leading to considerable physical self-discharge in ASSBs. This finding is corroborated by numerical and experimental investigations on ASSBs with varying SSE thicknesses. To achieve a self-discharge rate on par with commercial LIBs (<0.35% to ~2.5% for 1 month), the electronic conductivity of SSEs should be less than $10^{-12} \text{ S cm}^{-1}$. Unfortunately, none of the mainstream inorganic SSEs meets this criterion. Besides the electronic conductivity of SSEs, other factors, including interfacial reactions and ion redistribution within electrode materials, also lead to irreversible chemical self-discharge. Furthermore, by examining the mechanism behind physical and chemical self-discharge, multiscale strategies have been proposed to mitigate self-discharge of ASSBs, such as bandgap engineering, designing inorganic/organic composite electrolytes and interface engineering. In particular, interface engineering has proven effective in suppressing self-discharge, provided that an interphase with an electronic conductivity below $10^{-12} \text{ S cm}^{-1}$ is designed. This study provides valuable insights into the self-discharge of ASSBs and presents effective countermeasures to overcome this challenge.

Comparison of electronic conductivity of inorganic SSEs

The electronic conductivity of SSEs directly determines the physical self-discharge in ASSBs. Therefore, we first compared the electronic and ionic conductivities of inorganic SSEs commonly used in ASSBs. Original data are presented in Supplementary Table 1. As shown in Fig. 1a,b, the ionic conductivity of most sulfide, oxide and halide SSEs ranges from 10^{-4} to $10^{-2} \text{ S cm}^{-1}$ (refs. 6,21,22). Their electronic conductivity often falls between $10^{-8} \text{ S cm}^{-1}$ and $10^{-9} \text{ S cm}^{-1}$, considerably lower than their ionic conductivity ($\sim 10^{-4}$ – $10^{-2} \text{ S cm}^{-1}$). Due to this considerable difference, the electronic conductivity of SSEs has often been considered trivial in previous reports. However, contrary to conventional belief, the electronic conductivity of current inorganic SSEs is six to seven orders of magnitude higher compared with commercialized polymeric separators ($\sim 10^{-15}$ – $10^{-18} \text{ S cm}^{-1}$) and thin-film SSEs such as LiPON ($10^{-14} \text{ S cm}^{-1}$)²³ (Fig. 1c,d). The non-trivial electronic conductivity suggests that ASSBs using existing inorganic SSEs may experience considerable physical self-discharge, which may suspend their commercialization progress (Fig. 1e). Typically, commercial LIBs with polymeric separators have a low self-discharge rate (<0.35% to ~2.5% of the stored capacity over 1 month)²⁰. Similarly, commercialized thin-film batteries based on LiPON also exhibit a meagre self-discharge rate (<2% per year)²⁴. So far, it is unclear how severe the self-discharge in current ASSBs is. Therefore, quantitatively assessing self-discharge in ASSBs as a function of the electronic conductivity of inorganic SSEs is urgent and crucial for developing effective countermeasures.

Physical self-discharge affected by SSE thickness and electronic conductivity

To understand the physical self-discharge behaviour of ASSBs, we developed an electrochemical model including both the electron and ion

transport in the SSE. The electric current passing through the SSE, i_s (that is, self-discharge current), can be expressed as

$$i_s = -K_s \nabla \Phi_s, \quad (1)$$

where K_s and $\nabla \Phi_s$ denote the electric conductivity and the gradient of the electric potential in the SSE, respectively. The gradient $\nabla \Phi_s$ is roughly determined by the OCV and the thickness of the SSE when the self-discharge current and related charge-transfer overpotential is low and negligible. It is clearly seen that the physical self-discharge in ASSBs is primarily influenced by the electronic conductivity and thickness of the SSE, as well as the OCV of ASSBs. In the simulation, two types of ASSB configuration are analysed: laboratory-scale pellet-type model cells with a thick SSE (Fig. 2a) and practical sheet-type all-solid-state pouch cells with a thin SSE^{25,26} (ideally 20 μm ; Fig. 2b). Both configurations consist of three components, a composite cathode, SSE and lithium metal anode. The areal capacity and the OCV of the cells are set to 4 mAh cm^{-2} and 4.2 V, respectively. Additional details regarding the theoretical framework and numerical integration can be found in Methods.

Figure 2c illustrates the physical self-discharge behaviour of ASSBs with 1,000- μm SSEs. As the electronic conductivity of SSEs decreases from $9 \times 10^{-9} \text{ S cm}^{-1}$ to $1 \times 10^{-9} \text{ S cm}^{-1}$, the capacity retention ratio improves from 93.3% to 99.2%. The corresponding physical self-discharge over 1 month is minimal, ranging from only 0.8% to 6.7% (Fig. 2d). By sharp contrast, ASSBs with 20- μm SSEs exhibit notable physical self-discharge behaviour (Fig. 2e), with 100% self-discharge rate after 1 month when the electronic conductivity of SSEs exceeds $3 \times 10^{-9} \text{ S cm}^{-1}$ (Fig. 2f). To determine the critical electronic conductivity of SSEs that required to achieve a self-discharge rate of less than 5%, the self-discharge behaviour of ASSBs with 20- μm SSEs was simulated across a range of electronic conductivities, from $5 \times 10^{-9} \text{ S cm}^{-1}$ to $5 \times 10^{-12} \text{ S cm}^{-1}$ (Fig. 2g). The simulation reveals that, to maintain a self-discharge rate below 5% within 1 month, the electronic conductivity of SSEs must be lower than $5 \times 10^{-11} \text{ S cm}^{-1}$. Unfortunately, SSEs with such low electronic conductivity are rarely reported.

To examine the self-discharge behaviour of ASSBs, we fabricated ASSBs with different SSE thicknesses (1,000 μm versus 20 μm) via a solvent-free dry film process²⁷. Figure 3a shows the ASSBs ($\text{Li}_3\text{InCl}_6 @ \text{LiCoO}_2 / \text{Li}_6\text{PS}_4\text{Cl} / \text{In-Li}$) with a 1,000- μm $\text{Li}_6\text{PS}_4\text{Cl}$ layer. The commercial $\text{Li}_6\text{PS}_4\text{Cl}$ is selected as the SSE because of its moderate ionic conductivity ($2.01 \times 10^{-3} \text{ S cm}^{-1}$) and representative electronic conductivity ($3.65 \times 10^{-9} \text{ S cm}^{-1}$) at room temperature (Supplementary Figs. 1 and 2). Meanwhile, Li_3InCl_6 is used as the catholyte because of its decent room-temperature ionic conductivity ($1.486 \times 10^{-3} \text{ S cm}^{-1}$) and excellent interfacial chemical stability with LiCoO_2 (Supplementary Figs. 2 and 3). Figure 3b presents the cross-sectional images of the ASSBs with 1,000- μm $\text{Li}_6\text{PS}_4\text{Cl}$. The intimate contact at the electrode–electrolyte interface (Supplementary Fig. 4) and the uniform elemental distribution observed in energy-dispersive X-ray (EDX) mapping ensure the excellent electrochemical cycling stability of ASSBs (Supplementary Fig. 5). This stability lays the foundation for conducting reliable physical and chemical self-discharge analyses of ASSBs. As illustrated in Fig. 3c, ASSBs with a 1,000- μm $\text{Li}_6\text{PS}_4\text{Cl}$ layer experience a slow decay of voltage, with only minimal capacity loss of 3.9% observed within 1 month (Fig. 3d), which is comparable to conventional liquid LIBs (Supplementary Fig. 6). However, the energy density of ASSBs with such a thick SSE layer cannot go beyond 100 Wh kg^{-1} (Supplementary Fig. 7). To elucidate the contribution of physical and chemical self-discharge to the observed capacity loss, we assume that the capacity loss caused by physical self-discharge is fully recovered in the subsequent charging process, given that physical self-discharge primarily stems from the electronic conductivity of SSEs. As shown in Fig. 3e, 3.5% of the capacity loss is totally lost, indicating that chemical self-discharge is the predominant cause of the observed capacity loss under this circumstance.

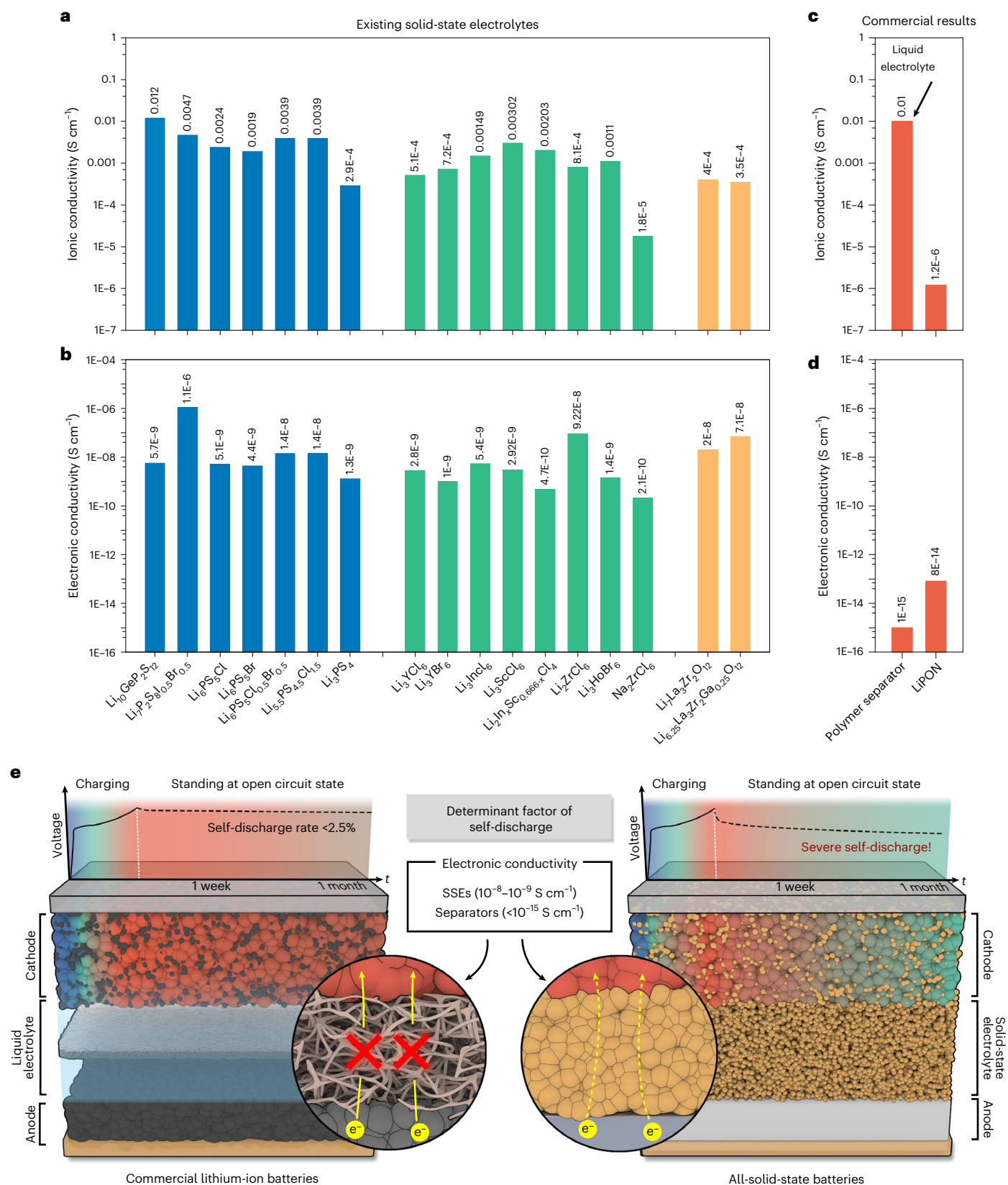


Fig. 1 | Comparative analysis of ionic and electronic conductivities of typical inorganic SSEs. a, b, Ionic conductivity (**a**) and electronic conductivity (**b**) of various inorganic SSEs. **c, d,** Ionic conductivity of commercial liquid electrolytes and LiPON (**c**) and electronic conductivity of polymeric separators and LiPON (**d**)

(original data and references can be found in Supplementary Table 1). These two materials have been successfully commercialized in LIBs and serve as references for this analysis. **e,** Schematic illustration of physical self-discharge of ASSBs caused by the non-negligible electronic conductivity of inorganic SSEs.

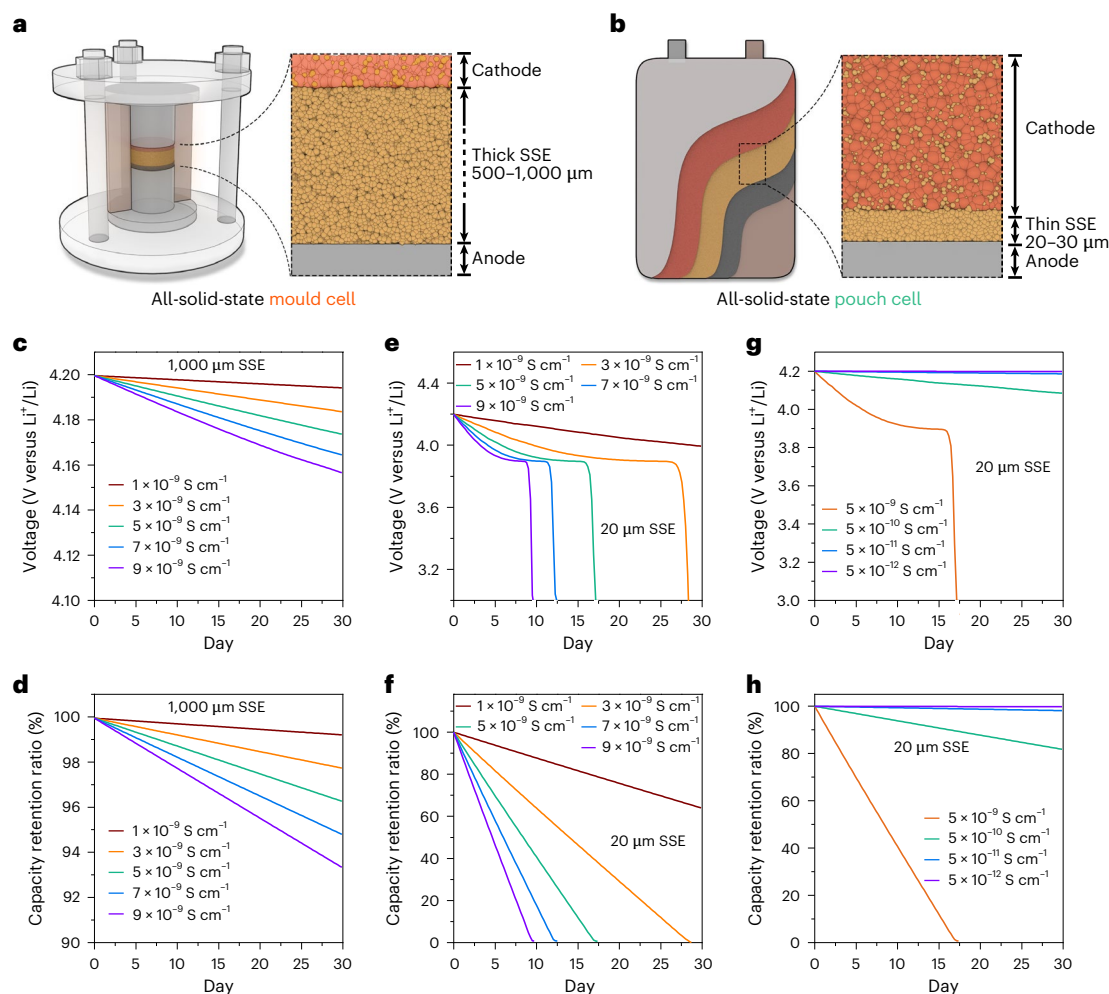


Fig. 2 | Simulation of the physical self-discharge behaviour of ASSBs as a function of the electronic conductivity and thickness of inorganic SSEs.

a, Schematic of a custom-made all-solid-state mould cell used in ASSB fundamental research, with an SSE thickness of approximately 1,000 μm. **b**, Schematic of all-solid-state pouch cells for practical application, where the SSE thickness is ideally 20 μm to ensure high energy density. **c,d**, Simulation of self-discharge behaviour of ASSBs with a 1,000-μm SSE with varying electronic conductivity

from $1 \times 10^{-9} \text{ S cm}^{-1}$ to $9 \times 10^{-9} \text{ S cm}^{-1}$ (**c**) and the corresponding capacity retention ratio time (**d**). **e,f**, Simulation of self-discharge behaviour of ASSBs with 20-μm SSE with varying electronic conductivity from $1 \times 10^{-9} \text{ S cm}^{-1}$ to $9 \times 10^{-9} \text{ S cm}^{-1}$ (**e**) and the corresponding capacity retention ratio over time (**f**). **g,h**, Simulation of self-discharge behaviour of ASSBs with 20 μm SSE with varying electronic conductivity from $5 \times 10^{-9} \text{ S cm}^{-1}$ to $5 \times 10^{-12} \text{ S cm}^{-1}$ (**g**) and the corresponding capacity retention ratio over time (**h**).

For a comparison, ASSBs with a 20-μm $\text{Li}_6\text{PS}_5\text{Cl}$ layer were also fabricated. As shown in Fig. 3g, 20-μm $\text{Li}_6\text{PS}_5\text{Cl}$ is sandwiched between the $\text{Li}_3\text{InCl}_6@\text{LiCoO}_2$ cathode layer and Li–In anode. More photos and scanning electron microscopy (SEM) images are included in Supplementary Fig. 8. The surface damage and detachment observed are caused by high-energy focused ion beam milling. The structural integrity of ASSBs was further corroborated through three-dimensional (3D) X-ray computed tomography (Supplementary Fig. 9). Surprisingly, ASSBs utilizing a 20-μm $\text{Li}_6\text{PS}_5\text{Cl}$ layer exhibit a rapid voltage decline within 1 month (Fig. 3h), with the capacity loss reaching up to 94.5% (Fig. 3i). Nevertheless, 83.2% of the loss capacity is recovered in the subsequent cycle (Fig. 3j), indicating that this capacity loss is mainly due to physical self-discharge, arising from the non-negligible electronic conductivity of $\text{Li}_6\text{PS}_5\text{Cl}$. The capacity loss caused by chemical self-discharge is 11.3%, higher than that of ASSBs with a 1,000-μm $\text{Li}_6\text{PS}_5\text{Cl}$ layer (3.5%), implying that the increased internal leakage current also exacerbates interfacial side reactions, leading to more pronounced irreversible chemical self-discharge. This conclusion is further supported by the slight increase in polarization voltage between the charge and discharge curves in the subsequent cycle (Fig. 3i). Supplementary Figs. 10 and 11 present unchanged X-ray

diffraction patterns and chemical states of $\text{Li}_3\text{InCl}_6@\text{LiCoO}_2$ cathode composites after self-discharge, respectively, providing further evidence that the observed self-discharge primarily arises from physical self-discharge resulting from the non-negligible electronic conductivity of $\text{Li}_6\text{PS}_5\text{Cl}$. Furthermore, the thickness-dependent self-discharge behaviour reveals that thinner SSE layers—particularly the 20-μm and 50-μm samples—exhibit more pronounced physical self-discharge, as indicated in Supplementary Fig. 12. In addition, the self-discharge-induced capacity loss follows a leakage-controlled scaling relation (Supplementary Fig. 13),

$$f_{\text{loss}} \approx \frac{\sigma_e \Delta V t}{q_A L}, \quad (2)$$

where f_{loss} is the fractional capacity loss after a storage time t , σ_e is the electronic conductivity, ΔV is the effective driving force, q_A is the areal capacity and L is the SSE thickness. This suggests that minimizing the SSE electronic conductivity and increasing SSE thickness are both effective ways to suppress capacity decay during storage. To prove physical self-discharge is ubiquitous in current ASSBs, a halide SSE (that is, Li_3YBr_6) membrane was used as a separator in ASSBs. Compared

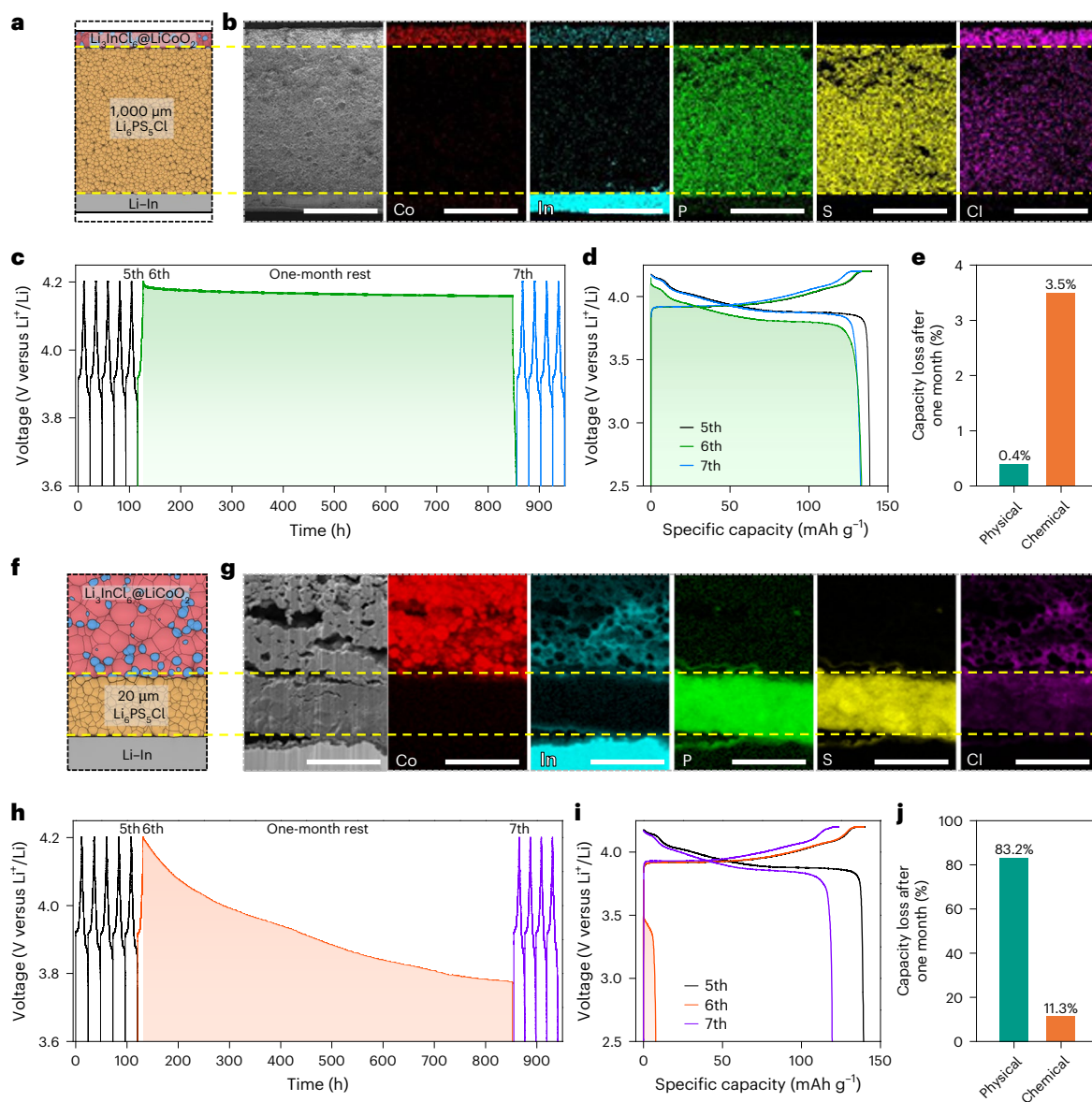


Fig. 3 | Experimental verification of self-discharge of ASSBs. **a**, Schematic illustration of an ASSB ($\text{Li}_3\text{InCl}_6 @ \text{LiCoO}_2 / \text{Li}_6\text{PS}_5\text{Cl} / \text{In-Li}$) with a 1,000- μm $\text{Li}_6\text{PS}_5\text{Cl}$ layer. **b**, SEM images to show the battery cross-section and its EDX mapping. Scale bars, 500 μm . **c**, The self-discharge behaviour of pellet-type ASSBs with 1,000- μm $\text{Li}_6\text{PS}_5\text{Cl}$. The battery was tested at 0.1 C (1 C = 140 mA g^{-1}). **d**, Charge-discharge curves of the ASSBs with 1,000- μm $\text{Li}_6\text{PS}_5\text{Cl}$ before and after 1-month ageing (0.1 C). **e**, The ratio of physical and chemical self-discharge in the capacity loss

observed in ASSBs with 1,000- μm $\text{Li}_6\text{PS}_5\text{Cl}$. **f**, Schematic illustration of an ASSB ($\text{Li}_3\text{InCl}_6 @ \text{LiCoO}_2 / \text{Li}_6\text{PS}_5\text{Cl} / \text{In-Li}$) with a 20- μm $\text{Li}_6\text{PS}_5\text{Cl}$ layer. **g**, SEM images of an enlarged area of the cross-section and its EDX mapping. Scale bars, 20 μm . **h**, The self-discharge behaviour of pellet-type ASSBs with a 20- μm SSE layer. **i**, Charge-discharge curves before and after 1-month ageing. **j**, The physical and chemical self-discharge ratio in the capacity loss observed in ASSBs with the 20- μm SSE.

with $\text{Li}_6\text{PS}_5\text{Cl}$, Li_3YBr_6 is a more stable SSE material against to metallic anodes (that is, Li, Li-In or In). Similarly, the ASSBs with a 20- μm Li_3YBr_6 membrane exhibit substantial leakage due to physical self-discharge. In addition, such physical self-discharge is also observed in the ASSBs using $\text{Li}_7\text{P}_3\text{S}_4$ as the separator (Supplementary Fig. 14).

The above experimental analyses convincingly demonstrate that the current electronic conductivity level of SSE in the range of 10^{-8} – 10^{-9} S cm^{-1} is not sufficiently low to eliminate the self-discharge of ASSBs. Therefore, reducing the electronic conductivity of SSE while maintaining high ionic conductivity is a crucial goal for future research. Considering that other factors, such as manufacturing defects, storage temperature, and humidity, also influence the self-discharge behaviour of ASSBs^{28–31}, we conservatively recommend decreasing the electronic conductivity of inorganic SSEs to approximately 10^{-12} S cm^{-1} for ASSBs

to achieve self-discharge rates on par with commercial LIBs, which typically have a monthly self-discharge rate of 2.5–0.35%²⁰.

Discussion

To minimize the self-discharge behaviour of ASSBs, multiscale strategies that can decrease the electronic conductivity of SSEs from their current level to the desired target (10^{-12} S cm^{-1}) should be explored in the future (Fig. 4a). At the atomic scale, enlarging the bandgap of SSEs by doping or substitution can increase the energy barrier for electron tunnelling, leading to low electronic conductivity. It is important to note that, when attempting to reduce the electronic conductivity of SSEs by adjusting the bandgap, care should be taken to ensure that the ionic conductivity of the SSE is not negatively impacted as a result³². According to Gorai et al., point defects such as vacancies, interstitials

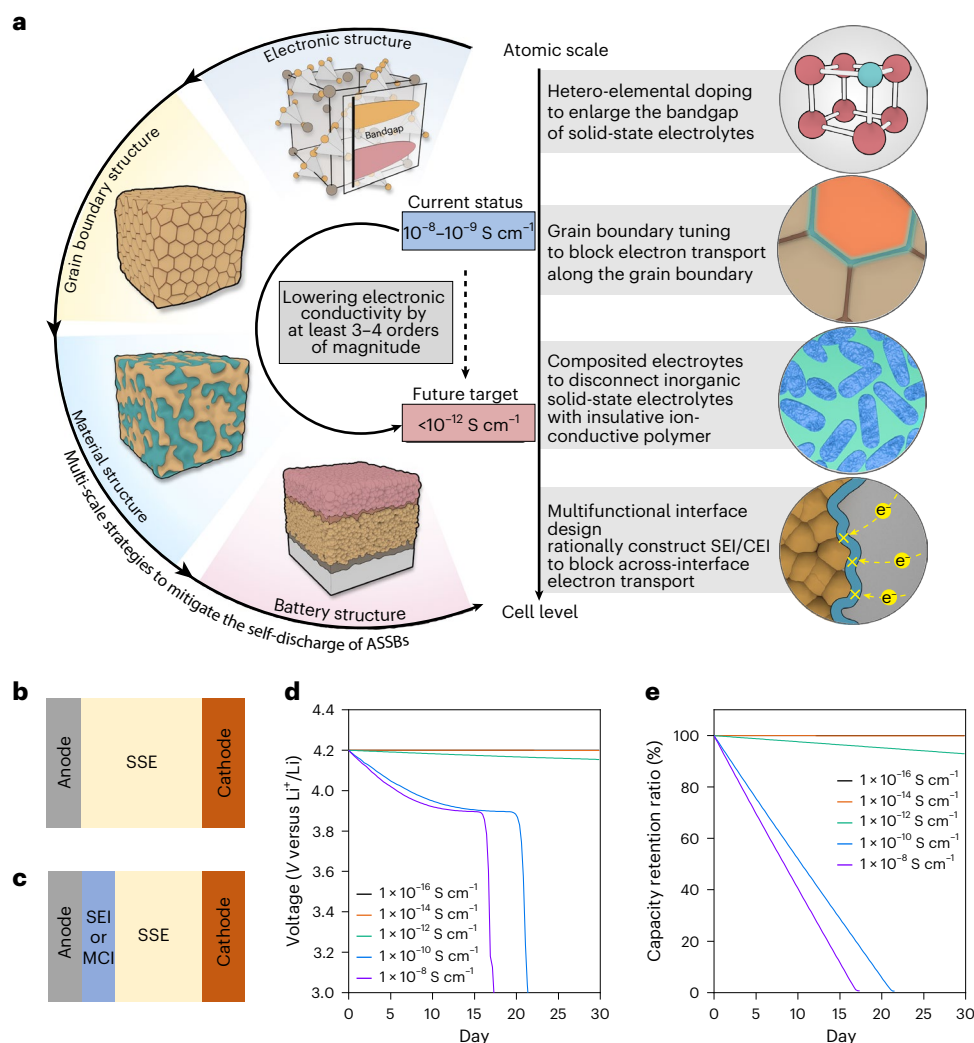


Fig. 4 | Multiscale strategies to mitigate physical self-discharge in ASSBs.

a, Multiscale strategies to reduce electronic conductivity of inorganic SSE from current status to a targeted electronic conductivity of less than 10^{-12} S cm $^{-1}$. SEI, solid electrolyte interphase; CEI, cathode electrolyte interphase. **b, c**, Schematic

illustration of ASSBs with **(b)** and without **(c)** interphase. **d, e**, Simulation of self-discharge behaviour of ASSBs with 20-μm SSE as well as a 0.1-μm interphase, which shows varying electronic conductivity from 1×10^{-8} S cm $^{-1}$ to 1×10^{-16} S cm $^{-1}$ (**d**) and the corresponding capacity retention ratio over time (**e**).

and antisites can enhance the ionic conductivity of inorganic SSEs, but they also increase their electronic conductivity³³. Therefore, it is important to pursue defect engineering strategies that can reduce the electronic conductivity of SSEs while maintaining high ionic conductivity. In addition, investigating the relationship between the electronic energy bandgap and the electronic and ionic conductivities of inorganic SSEs is also essential for guiding future research efforts. At grain boundaries, lithium dendrites can readily form due to their higher electronic conductivity compared with the bulk material^{28,34,35}, often leading to soft short circuits and battery failure. In view of this, tuning the grain boundaries to reduce electron transport is also crucial for suppressing self-discharge. At the microscale, it is worth considering the development of composite SSEs. Polymeric separators have very low electronic conductivity (10^{-15} – 10^{-18} S cm $^{-1}$)^{36–39}. Therefore, by combining an inorganic SSE with polymeric materials, the electronic conductivity can be reduced by breaking up the electron pathways along the inorganic SSE^{40,41}. However, if the polymer has no ionic conductivity, the ion conductivity of the SSE will be reduced as well. Using a single-ion conducting polymer may be a promising solution⁴². At the interface, we can design a multifunctional interface to allow only ion transport while blocking electrons^{43,44}, which is not only helpful in reducing self-discharge but also in suppressing interfacial side reactions and dendrite growth⁴⁵.

To demonstrate the impact of interface engineering on mitigating self-discharge, we simulated a 20-μm SSE combined with a 0.1-μm coating layer possessing variable electronic conductivities ranging from 10^{-8} S cm $^{-1}$ to 10^{-12} S cm $^{-1}$ (Fig. 4b,c). When the interphase layer exhibits an electronic conductivity below 10^{-12} S cm $^{-1}$, the self-discharge behaviour of ASSBs is significantly suppressed (Fig. 4d), resulting in a capacity retention ratio exceeding 93% (Fig. 4e).

In summary, we investigated the self-discharge behaviour of ASSBs by numerical and experimental analyses and quantified the contribution of physical and chemical factors to the observed capacity loss. Our findings indicate that the current electronic conductivity level of inorganic SSEs, which is typically in the range of 10^{-8} – 10^{-9} S cm $^{-1}$, leads to significant physical self-discharge in ASSBs, particularly when the SSE layer thickness is on the order of tens of micrometres. Undoubtedly, this considerable self-discharge rate of ASSBs limits their wide adoption in large-scale energy storage systems, such as EVs and grid-scale energy storage systems and causes challenges in battery management system design and implementation. To achieve a self-discharge rate comparable to commercial LIBs ($<0.35\%$ to $\sim 2.5\%$ over 1 month), it is necessary to reduce the electronic conductivity of SSEs to approximately 10^{-12} S cm $^{-1}$. Unfortunately, none of the mainstream inorganic SSEs used in ASSBs meets this requirement. To suppress the physical

self-discharge of ASSBs, multiscale strategies have been proposed, such as bandgap engineering and designing inorganic/organic composite electrolytes. In particular, interface engineering has been shown to be effective, provided that an interphase with an electronic conductivity below $10^{-12} \text{ S cm}^{-1}$ is designed. This study provides valuable insights into ASSB research and will inform their future design and development.

Methods

Numerical simulation

A pseudo-two-dimensional model was employed to simulate the electrochemical behaviour of the composite cathode, where LiCoO_2 particles are uniformly dispersed in the Li_3InCl_6 matrix. The theoretical framework of the electrochemical model is outlined as follows. The electric current density $\mathbf{i}_l$ in the SSE due to ion transport can be described by the Nernst–Planck equation,

$$\mathbf{i}_l = (-K_l \nabla \Phi_l) + \frac{2K_l RT}{F} \left(1 + \frac{\partial \ln f}{\partial \ln C_l} \right) (1 - t_+) \nabla \ln C_l, \quad (3)$$

where Φ_l is the potential in the SSE, K_l is the ionic conductivity of the SSE, C_l is the Li^+ concentration in the SSE, R is the gas constant, T is the temperature, F is the Faraday's constant, f is the mean molar activity coefficient of the electrolyte and t_+ is the transference number of Li^+ . For the single-ion conductor such as $\text{Li}_6\text{PS}_5\text{Cl}$ and Li_3InCl_6 with the t_+ of 1.0. Equation (3) simplifies to

$$\mathbf{i}_l = -K_l \nabla \Phi_l. \quad (4)$$

Charge conservation in the electrolyte mandates that the field of $\mathbf{i}_l$ be divergence-free:

$$\nabla \cdot \mathbf{i}_l = 0 \quad (5)$$

Because the electric conductivity of SSE is non-negligible, the electric current density $\mathbf{i}_s$ due to electron transport in the SSE is described by Ohm's law,

$$\mathbf{i}_s = -K_s \nabla \Phi_s, \quad (6)$$

where K_s and Φ_s represent the electric conductivity and electric potential of the SSE, respectively. For charge conservation, the field of $\mathbf{i}_e$ must satisfy

$$\nabla \cdot \mathbf{i}_s = 0. \quad (7)$$

Charge-transfer reactions are governed by the overpotential η . The charge-transfer rate i is related to the overpotential by the Butler–Volmer equation:

$$i = i_0 \left(\exp \left(\frac{\alpha_a F \eta}{RT} \right) - \exp \left(-\frac{\alpha_c F \eta}{RT} \right) \right), \quad (8)$$

where α_a and α_c are the anodic and cathodic transfer coefficients, respectively. The overpotential at the $\text{LiCoO}_2/\text{Li}_3\text{InCl}_6$ interface is defined as

$$\eta = \Phi_s - \Phi_l - \Phi_{\text{eq}}, \quad (9)$$

where Φ_{eq} is the equilibrium potential for the Li reaction in the active material. The exchange current density i_0 is given as

$$i_0 = F(k_c)^{\alpha_a} (k_a)^{\alpha_c} (C_{s,\text{max}} - C_s)^{\alpha_a} (C_s)^{\alpha_c}, \text{ at the } \text{LiCoO}_2/\text{Li}_3\text{InCl}_6 \text{ interface}, \quad (10)$$

where k_c and k_a are the rate constants for the anodic and cathodic reactions, C_s is the bulk Li^+ concentration in the active material, and $C_{s,\text{max}}$ is the maximum Li^+ concentration in the LiCoO_2 particles.

For Li plating/stripping at the Li/SSE interface, the overpotential is defined as

$$\eta = \Phi_s - \Phi_l. \quad (11)$$

The exchange current density i_0 is expressed as

$$i_0 = i_{0,\text{plat}}. \quad (12)$$

The mass transport of Li^+ within the LiCoO_2 particles is governed by

$$\frac{\partial C_s}{\partial t} + \nabla \cdot \mathbf{J}_s = 0 \quad (13)$$

$$\mathbf{J}_s = -D_s \nabla C_s, \quad (14)$$

where $\mathbf{J}_s$ represents the flux of Li^+ due to solid-state diffusion in the LiCoO_2 particles, and D_s is the Li^+ diffusivity in the particles. The effective transport parameters of the composite cathode are corrected via the Bruggeman equation:

$$D_{\text{eff}} = \varepsilon^{1.5} D_{\text{bulk}}, \quad (15)$$

where D_{eff} is the effective transport parameters (for example, electric conductivity and ion conductivity), D_{bulk} is the intrinsic transport parameter of the SSE, and ε is the volume fraction of the SSE in the composite cathode. The governing equations are solved numerically using the electrochemical module in COMSOL Multiphysics. The geometry and boundary conditions of the numerical model are illustrated in Supplementary Fig. 15 and Supplementary Table 2.

Preparation of materials

$\text{Li}_6\text{PS}_5\text{Cl}$ was supplied by MSE supplier LLC. Both were stored in an Ar-filled glove box and used as received without any treatment. Li_3InCl_6 was synthesized as described in our previous paper⁴⁶. Stoichiometric amounts of LiCl (Sigma-Aldrich, 99.9%) and InCl_3 (Alfa Aesar, 99.99%) were dissolved in deionized water in the ambient environment. The solution was dried in a 60 °C oven to obtain $\text{Li}_3\text{InCl}_6 \cdot x\text{H}_2\text{O}$ hydrate, which was further heated at 200 °C under vacuum. LiCoO_2 was purchased from. LiCoO_2 (Alfa Aesar, 99.5%, $D_{50} = 5 \mu\text{m}$) is coated with 5 nm LiNbO_3 by atomic layer deposition to guarantee interfacial stability, as described in our previous paper⁴⁷. Lithium *tert*-butoxide (LiO^tBu , $(\text{CH}_3)_3\text{COLi}$, Alfa Aesar, >99.9%) and niobium ethoxide ($\text{Nb}(\text{OEt})_5$, $\text{Et} = -\text{CH}_2\text{CH}_3$, Strem Chemicals, >99.9%) were used for Li and Nb sources, respectively. All of the depositions were performed at a temperature of 235 °C. $\text{Li}_7\text{P}_3\text{S}_4$ was obtained from HF-Kejing and used as received without any treatment. Li_3YBr_6 was synthesized using a solid-state reaction route by ball-milling a stoichiometric mixture of LiCl (Sigma-Aldrich, 99.9%) and BrCl_3 (Alfa Aesar, 99.99%) for 15 h at 500 rpm, followed by calcination at 500 °C for 5 h.

Electronic and ionic conductivity measurements

The ionic conductivity of SSEs was determined by a.c. electrochemical impedance spectroscopy with 100 mV constant voltage within a frequency range of 7 MHz to 100 mHz using a VMP3 potentiostat/galvanostat (BioLogic). Typically, -100–120 mg SSE power was placed in a hollow polyether ether ketone (PEEK) cylindrical die and pressed into a pellet with a diameter of 10 mm by the hydraulic press at 3 tons for 1 min in an Ar-filled glove box ($\text{H}_2\text{O} < 0.01 \text{ ppm}$, $\text{O}_2 < 0.01 \text{ ppm}$). Carbon was added to improve electronic contact. The electronic conductivities of the samples were measured by d.c. polarization tests with externally applied voltages of from 0.1 V to 0.5 V for 2 h to reach a steady voltage.

Pellet-type ASSB fabrication and electrochemical measurements

Pellet-type ASSBs with a configuration of $\text{Li}_3\text{InCl}_6/\text{LiCoO}_2/\text{Li}_6\text{PS}_5\text{Cl}/\text{In-Li}$ were assembled in an ultrapure Ar-filled glove box ($\text{H}_2\text{O} < 0.01 \text{ ppm}$,

$O_2 < 0.01$ ppm). A dry-film process was used to prepare cathode composite sheets and SSE membranes with different thicknesses²⁷. First, 70 wt% $LiCoO_2$ with 30 wt% Li_3InCl_6 were hand-milled in an agate mortar for ~10–15 min to ensure their uniform mixing. Then, 1 wt% polytetrafluoroethylene (supplied by GLABAT Solid-State Battery) was added. A cathode composite dough was obtained by shear mixing. A 25- μ m stainless-steel foil was used as the substrate, and the dough was rolled and pressed into a 100- μ m cathode composite film, which was then cut using a hollow steel punch (3/8 inch). Second, Li_6PS_5Cl powder was mixed with 1 wt% polytetrafluoroethylene binder and rolled into a film on a stainless-steel substrate. The calendaring process was used to improve its uniformity in thickness. To obtain 1,000- μ m film, multilayer Li_6PS_5Cl membranes were stacked together; 20- μ m Li_6PS_5Cl membrane was obtained via a repeated calendaring process. Third, a 20- μ m Li foil (MSE Suppliers) and a 50- μ m indium foil (99.99%) were pressed together to form an In–Li thin layer. To assemble the model cell, the Li_6PS_5Cl membrane was pressed at 380 MPa first. Then cathode composite membrane (~12–15 mg, ~100 μ m) was pressed on one side of SSE under low pressure. Lastly, the Li–In alloy anode was attached to the other side of the Li_6PS_5Cl membrane. The cell was placed into a custom-made stainless-steel case with air-tight sealing. A constant pressure of 10 MPa was applied during electrochemical measurement. Galvanostatic cycling of the ASSBs was carried out with NEWARE battery analysers with a voltage range of ~0–5 V and a current range of 10 mA at room temperature. ASSBs were charged to 4.2 V (versus Li^+/Li) with a constant current of 0.1 C (1 C = 140 mA g⁻¹) and then charged by a constant voltage of 4.2 V until the current reached 0.01 C, followed by a discharging process to 2.5 V (versus Li^+/Li) using a constant current of 0.1 C. To analyse the self-discharge behaviour, ASSBs were stably cycled for five cycles at 0.1 C, charged to 4.2 V and then aged for 1 month. After that, ASSBs were discharged to 2.5 V, followed by the galvanostatic charging/discharging process at 0.1 C.

Characterizations

X-ray diffraction patterns were collected using a Bruker AXS D8 Advance diffractometer with Cu K α radiation ($\lambda = 1.54178$ Å) at room temperature. The powder sample was loaded in an Ar-filled glove box and protected with a Kapton film. The cross-section morphology and its EDX mapping were examined using a Hitachi S-4800 field-emission scanning electron microscope equipped with EDX. To observe the cross-section of ASSBs with 20- μ m Li_6PS_5Cl , the sample was first cut with scissors and then milled by the focused ion beam (LEO (Zeiss) 1540XB microscope). X-ray microscopy imaging was performed using a 3D X-ray microscope (Xradia 515 Versa, Zeiss) operating at an accelerating voltage of 140 kV and power of 10 W. A LE6 source filter was used to reduce beam-hardening artefacts. A total of 1,401 projection images were obtained over 180°, each one with an exposure time of 2 s. The pixel size of the images was 2 μ m. Interfacial reaction energy data were obtained from the Materials Project database.

Data availability

All data that supported the findings and the conclusion are present in the analysis and its Supplementary Information.

References

- Kato, Y. et al. High-power all-solid-state batteries using sulfide superionic conductors. *Nat. Energy* **1**, 16030 (2016).
- Lee, Y.-G. et al. High-energy long-cycling all-solid-state lithium metal batteries enabled by silver–carbon composite anodes. *Nat. Energy* **5**, 299–308 (2020).
- Tan, D. H. S., Banerjee, A., Chen, Z. & Meng, Y. S. From nanoscale interface characterization to sustainable energy storage using all-solid-state batteries. *Nat. Nanotech.* **15**, 170–180 (2020).
- Fan, L.-Z., He, H. & Nan, C.-W. Tailoring inorganic–polymer composites for the mass production of solid-state batteries. *Nat. Rev. Mater.* **6**, 1003–1019 (2021).
- Manthiram, A., Yu, X. & Wang, S. Lithium battery chemistries enabled by solid-state electrolytes. *Nat. Rev. Mater.* **2**, 16103 (2017).
- Kamaya, N. et al. A lithium superionic conductor. *Nat. Mater.* **10**, 682–686 (2011).
- Han, X. et al. Negating interfacial impedance in garnet-based solid-state Li metal batteries. *Nat. Mater.* **16**, 572–579 (2017).
- Wang, C. et al. Garnet-type solid-state electrolytes: materials, interfaces, and batteries. *Chem. Rev.* **120**, 4257–4300 (2020).
- Liang, J., Li, X., Adair, K. R. & Sun, X. Metal halide superionic conductor for all-solid-state batteries. *Acc. Chem. Res.* **54**, 1023–1033 (2021).
- Kwak, H. et al. Emerging halide superionic conductors for all-solid-state batteries: design, synthesis, and practical applications. *ACS Energy Lett.* **7**, 1776–1805 (2022).
- Asano, T. et al. Solid halide electrolytes with high lithium-ion conductivity for application in 4 V class bulk-type all-solid-state batteries. *Adv. Mater.* **0**, 1803075 (2018).
- Li, X. et al. Progress and perspectives on halide lithium conductors for all-solid-state lithium batteries. *Energy Environ. Sci.* **13**, 1429–1461 (2020).
- Wang, C., Liang, J., Kim, J. T. & Sun, X. Prospects of halide-based all-solid-state batteries: from material design to practical application. *Sci. Adv.* **8**, eadc9516 (2022).
- Xia, W. et al. Antiperovskite electrolytes for solid-state batteries. *Chem. Rev.* **122**, 3763–3819 (2022).
- Matsuo, M. & Orimo, S.-i. Lithium fast-ionic conduction in complex hydrides: review and prospects. *Adv. Energy Mater.* **1**, 161–172 (2011).
- Kim, S. et al. A complex hydride lithium superionic conductor for high-energy-density all-solid-state lithium metal batteries. *Nat. Commun.* **10**, 1081 (2019).
- Xu, K. Nonaqueous liquid electrolytes for lithium-based rechargeable batteries. *Chem. Rev.* **104**, 4303–4418 (2004).
- Tan, D. H. S. et al. Carbon-free high-loading silicon anodes enabled by sulfide solid electrolytes. *Science* **373**, 1494–1499 (2021).
- Zhou, L. et al. High areal capacity, long cycle life 4 V ceramic all-solid-state Li-ion batteries enabled by chloride solid electrolytes. *Nat. Energy* **7**, 83–93 (2022).
- Seong, W. M. et al. Abnormal self-discharge in lithium-ion batteries. *Energy Environ. Sci.* **11**, 970–978 (2018).
- Zhou, L., Assoud, A., Zhang, Q., Wu, X. & Nazar, L. F. A new family of argyrodite thioantimonate lithium superionic conductors. *J. Am. Chem. Soc.* **141**, 19002–19013 <https://doi.org/10.1021/jacs.9b08357> (2019).
- Liang, J. et al. $Li_{10}Ge(P_{1-x}Sb_x)_2S_{12}$ lithium-ion conductors with enhanced atmospheric stability. *Chem. Mater.* **32**, 2664–2672 (2020).
- Le Van-Jodin, L., Ducroquet, F., Sabary, F. & Chevalier, I. Dielectric properties, conductivity and Li^+ ion motion in LiPON thin films. *Solid State Ion.* **253**, 151–156 (2013).
- Dudney, N. J. Solid-state thin-film rechargeable batteries. *Mater. Sci. Eng. B* **116**, 245–249 (2005).
- Randau, S. et al. Benchmarking the performance of all-solid-state lithium batteries. *Nat. Energy* **5**, 259–270 (2020).
- Albertus, P., Babinec, S., Litzelman, S. & Newman, A. Status and challenges in enabling the lithium metal electrode for high-energy and low-cost rechargeable batteries. *Nat. Energy* **3**, 16–21 (2018).
- Wang, C. et al. Solvent-free approach for interweaving freestanding and ultrathin inorganic solid electrolyte membranes. *ACS Energy Lett.* **7**, 410–416 (2022).
- Han, F. et al. High electronic conductivity as the origin of lithium dendrite formation within solid electrolytes. *Nat. Energy* **4**, 187–196 (2019).

29. Song, Y. et al. Probing into the origin of an electronic conductivity surge in a garnet solid-state electrolyte. *J. Mater. Chem. A* **7**, 22898–22902 (2019).
30. Yoon, K. et al. Detrimental effect of high-temperature storage on sulfide-based all-solid-state batteries. *Appl. Phys. Rev.* **9**, 031403 (2022).
31. McBrayer, J. D. et al. Calendar aging of silicon-containing batteries. *Nat. Energy* **6**, 866–872 (2021).
32. Wang, X., Chen, J., Mao, Z. & Wang, D. Effective resistance to dendrite growth of NASICON solid electrolyte with lower electronic conductivity. *J. Chem. Eng.* **427**, 130899 (2022).
33. Gorai, P., Famprikis, T., Singh, B., Stevanović, V. & Canepa, P. Devil is in the defects: electronic conductivity in solid electrolytes. *Chem. Mater.* **33**, 7484–7498 (2021).
34. Wang, C. et al. Identifying soft breakdown in all-solid-state lithium battery. *Joule* **6**, 1770–1781 (2022).
35. Liu, X. et al. Local electronic structure variation resulting in Li ‘filament’ formation within solid electrolytes. *Nat. Mater.* **20**, 1485–1490 (2021).
36. Nestler, T. et al. Separators-technology review: ceramic based separators for secondary batteries. *AIP Conf. Proc.* **1597**, 155–184 (2014).
37. Ji, Y. & Jiang, Y. Increasing the electrical conductivity of poly(vinylidene fluoride) by KrF excimer laser irradiation. *Appl. Phys. Lett.* **89**, 221103 (2006).
38. Foss, R. A. & Dannhauser, W. Electrical conductivity of polypropylene. *J. Appl. Polym. Sci.* **7**, 1015–1022 (1963).
39. Adamec, V. & Calderwood, J. H. On the determination of electrical conductivity in polyethylene. *J. Phys. D* **14**, 1487–1494 (1981).
40. Dixit, M. B. et al. Nanoscale mapping of extrinsic interfaces in hybrid solid electrolytes. *Joule* **4**, 207–221 (2020).
41. Liu, W. et al. Enhancing ionic conductivity in composite polymer electrolytes with well-aligned ceramic nanowires. *Nat. Energy* **2**, 17035 (2017).
42. Fan, L.-Z., He, H. & Nan, C.-W. Tailoring inorganic–polymer composites for the mass production of solid-state batteries. *Nat. Rev. Mater.* **6**, 1003–1019 <https://doi.org/10.1038/s41578-021-00320-0> (2021).
43. Tu, Q., Shi, T., Chakravarthy, S. & Ceder, G. Understanding metal propagation in solid electrolytes due to mixed ionic-electronic conduction. *Matter* **4**, 3248–3268 (2021).
44. Zhang, W. et al. Tuning the LUMO energy of an organic interphase to stabilize lithium metal batteries. *ACS Energy Lett.* **4**, 644–650 (2019).
45. Shen, Z. et al. Tuning the interfacial electronic conductivity by artificial electron tunneling barriers for practical lithium metal batteries. *Nano Lett.* **20**, 6606–6613 (2020).
46. Li, X. et al. Water-mediated synthesis of a superionic halide solid electrolyte. *Angew. Chem. Int. Ed.* **131**, 16579–16584 (2019).
47. Wang, C. et al. Manipulating interfacial nanostructure to achieve high-performance all-solid-state lithium-ion batteries. *Small Methods* **3**, 1900261 (2019).

Author contributions

C.W. conceived the idea, conducted characterization and analyses, and wrote the paper. R.X. performed the numerical simulation. Y.Z. helped with figures and data analysis. S.L., H.S., H.R. and J.T.K. helped with electrochemical analyses. B.X. and L.J. performed the 3D X-ray computed tomography analysis. H.H., J.T. and X.S. supervised the whole project and revised the paper before submission.

Funding

This work is funded by the National Natural Science Foundation of China (grant nos 92572112, W2441017 and 22409103), the National Key R&D Program of China (grant no. 2025YFF0523000), the Natural Science Foundation of Zhejiang Province (grant no. LZ26E020006), the Commanding Heights of Science and Technology of Chinese Academy of Sciences (LDES15 0000) and the ‘Innovation Yongjiang 2035’ Key R&D Program (grant nos 2024Z040 and 2025Z063).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-026-02090-x>.

Correspondence and requests for materials should be addressed to Changhong Wang or Xueliang Sun.

Peer review information *Nature Energy* thanks Yuping Wu and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher’s note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026